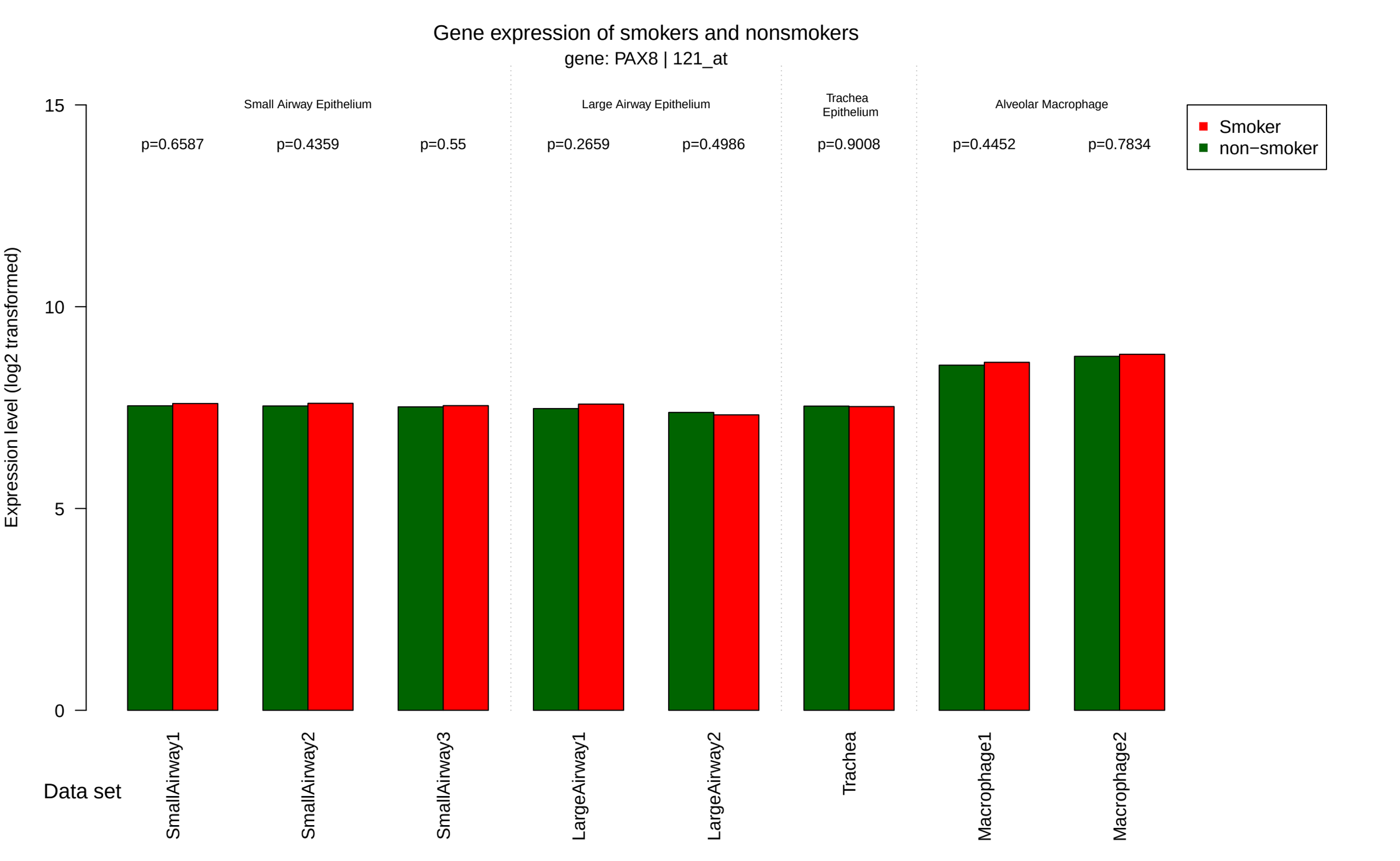
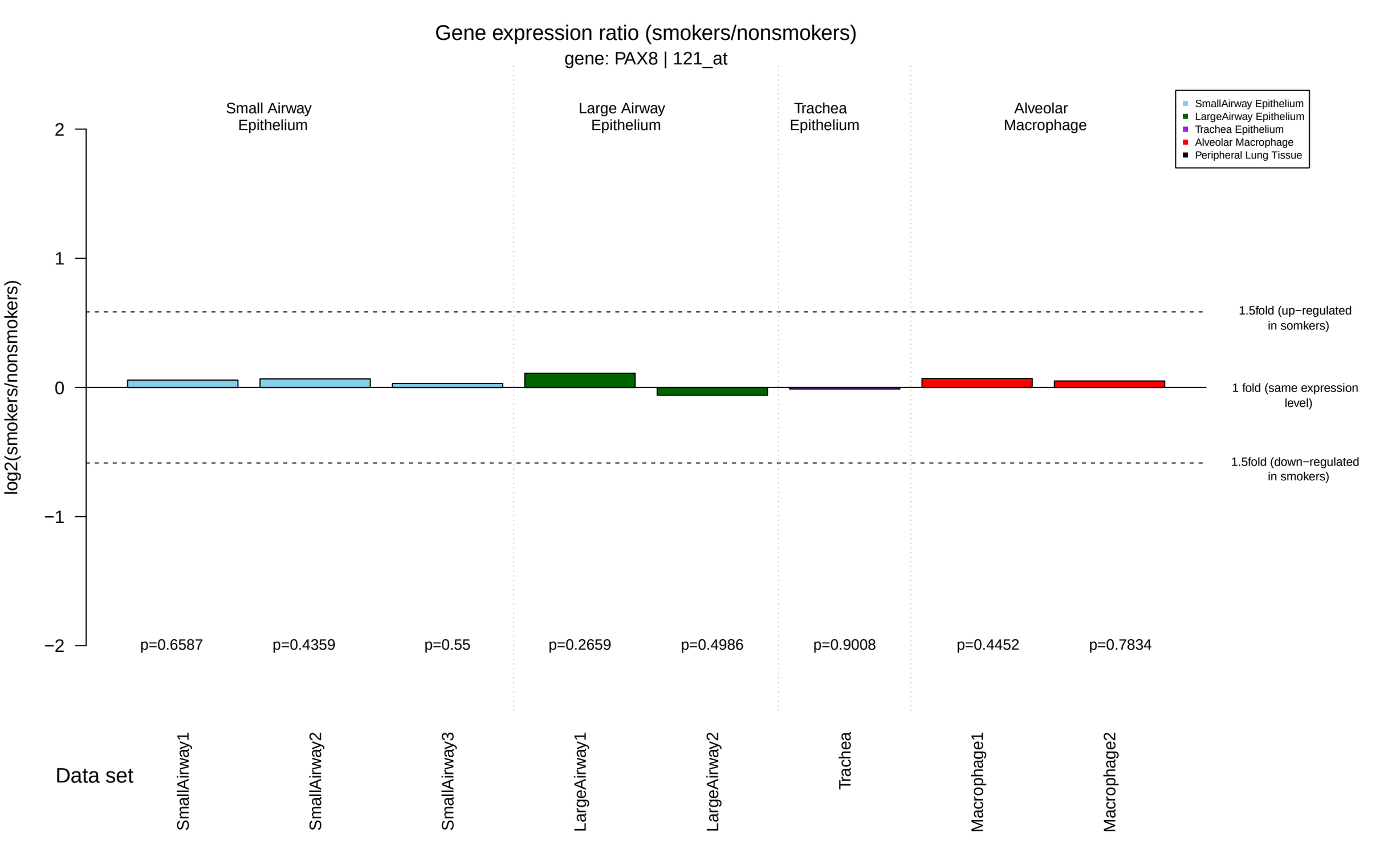
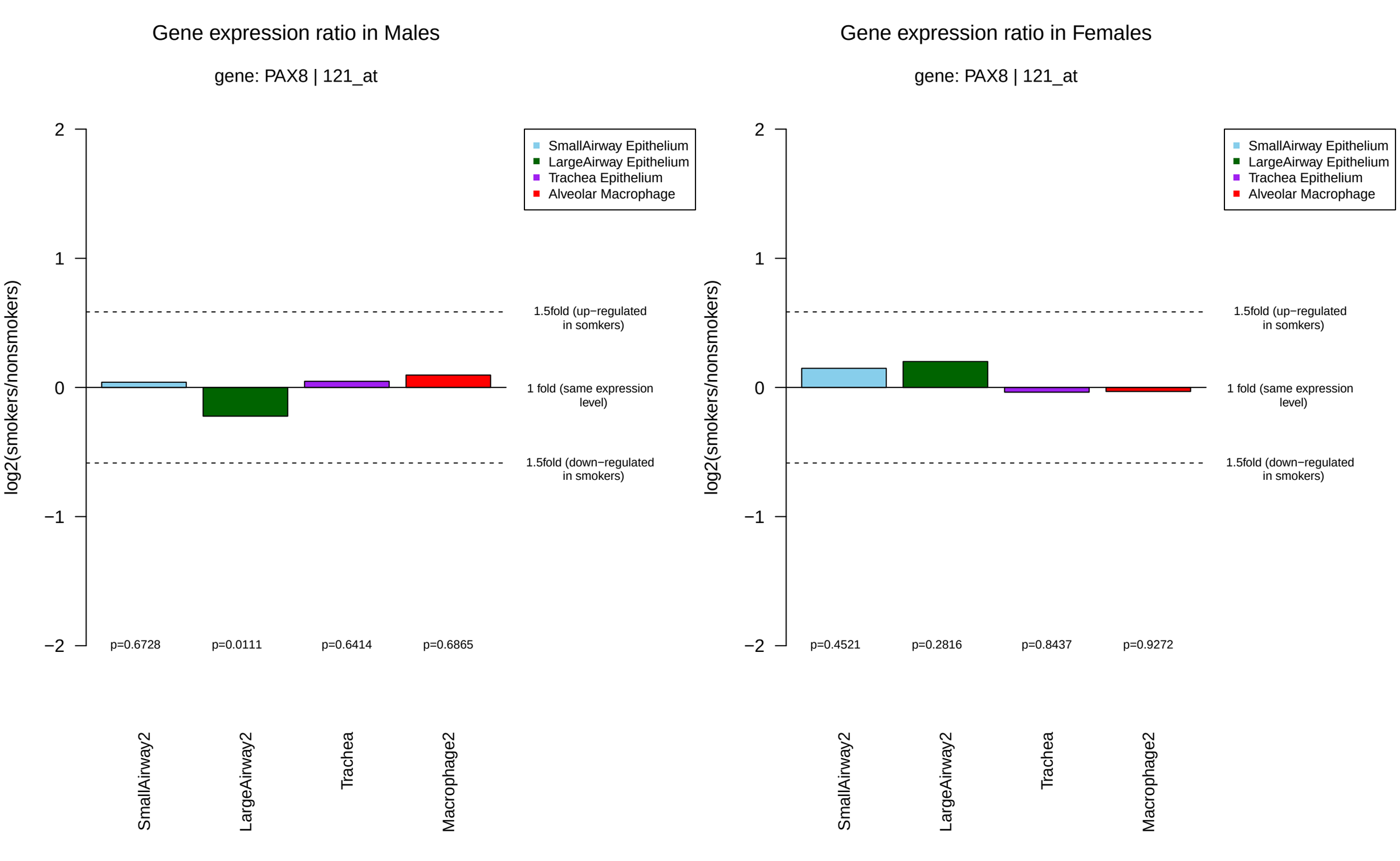
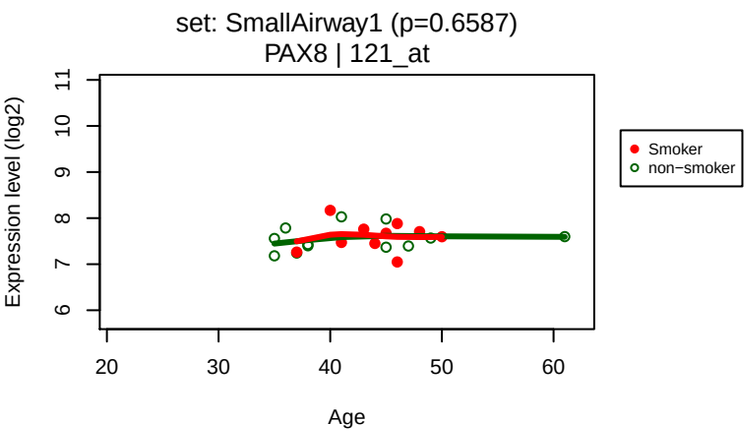
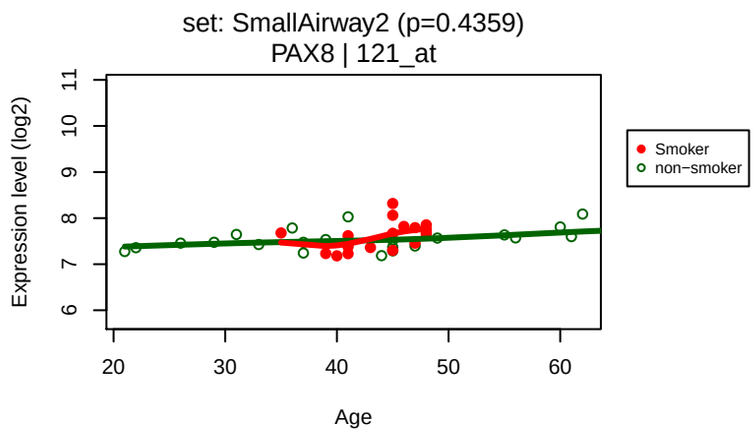
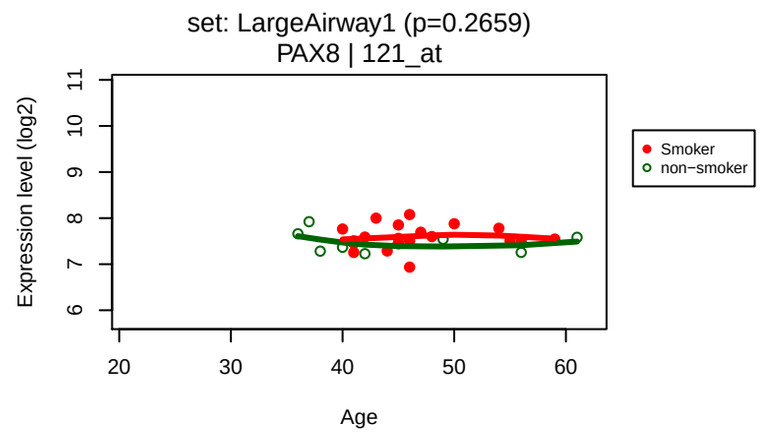

Gene expression of smokers and nonsmokers
gene: PAX8 | 121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6587
p=0.4359
p=0.55
p=0.2659
p=0.4986
p=0.9008
p=0.4452
p=0.7834
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

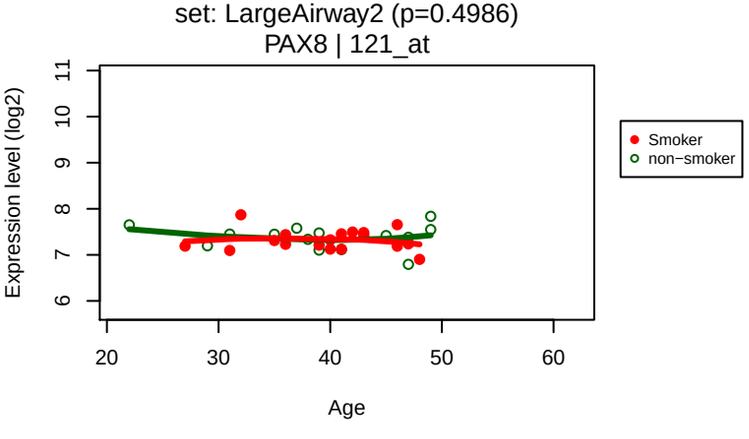

Gene expression ratio (smokers/nonsmokers)
gene: PAX8 | 121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6587
p=0.4359
p=0.55
p=0.2659
p=0.4986
p=0.9008
p=0.4452
p=0.7834
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

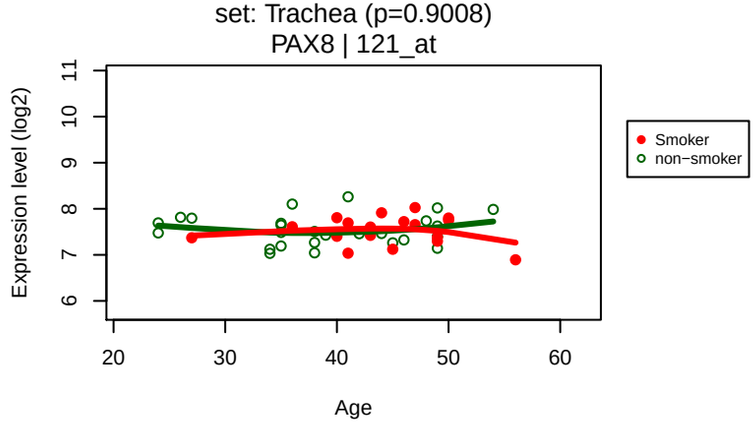

Gene expression ratio in Males
gene: PAX8 | 121_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6728
p=0.0111
p=0.6414
p=0.6865
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX8 | 121_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4521
p=0.2816
p=0.8437
p=0.9272
SmallAirway2
LargeAirway2
Trachea
Macrophage2

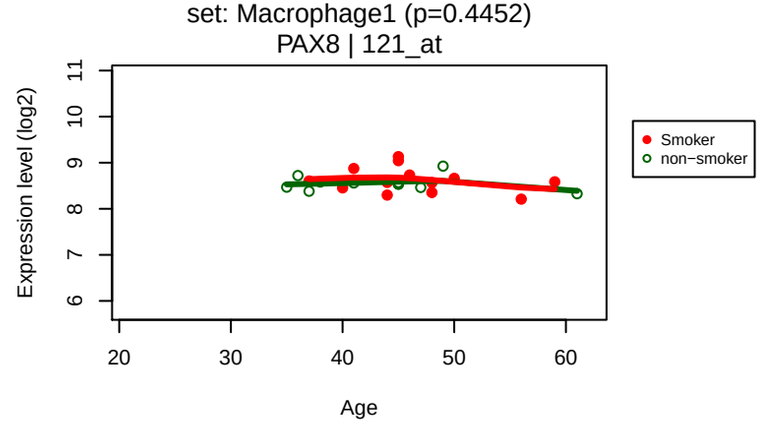

set: SmallAirway1 (p=0.6587)
PAX8 | 121_at
Expression level (log2)
Age
Smoker
non−smoker

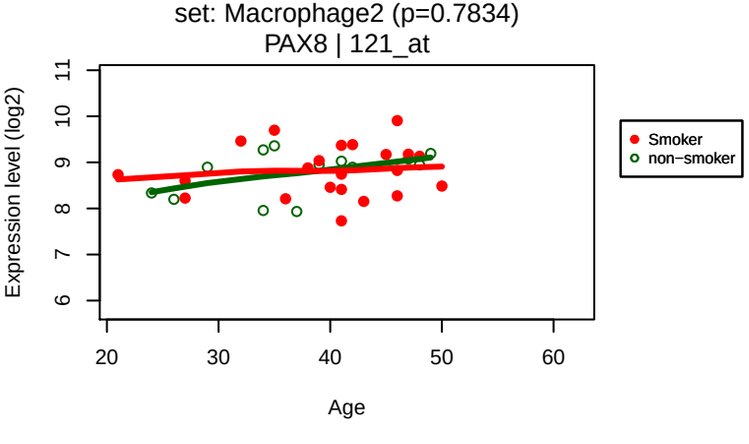

set: SmallAirway2 (p=0.4359)
PAX8 | 121_at
Expression level (log2)
Age
Smoker
non−smoker

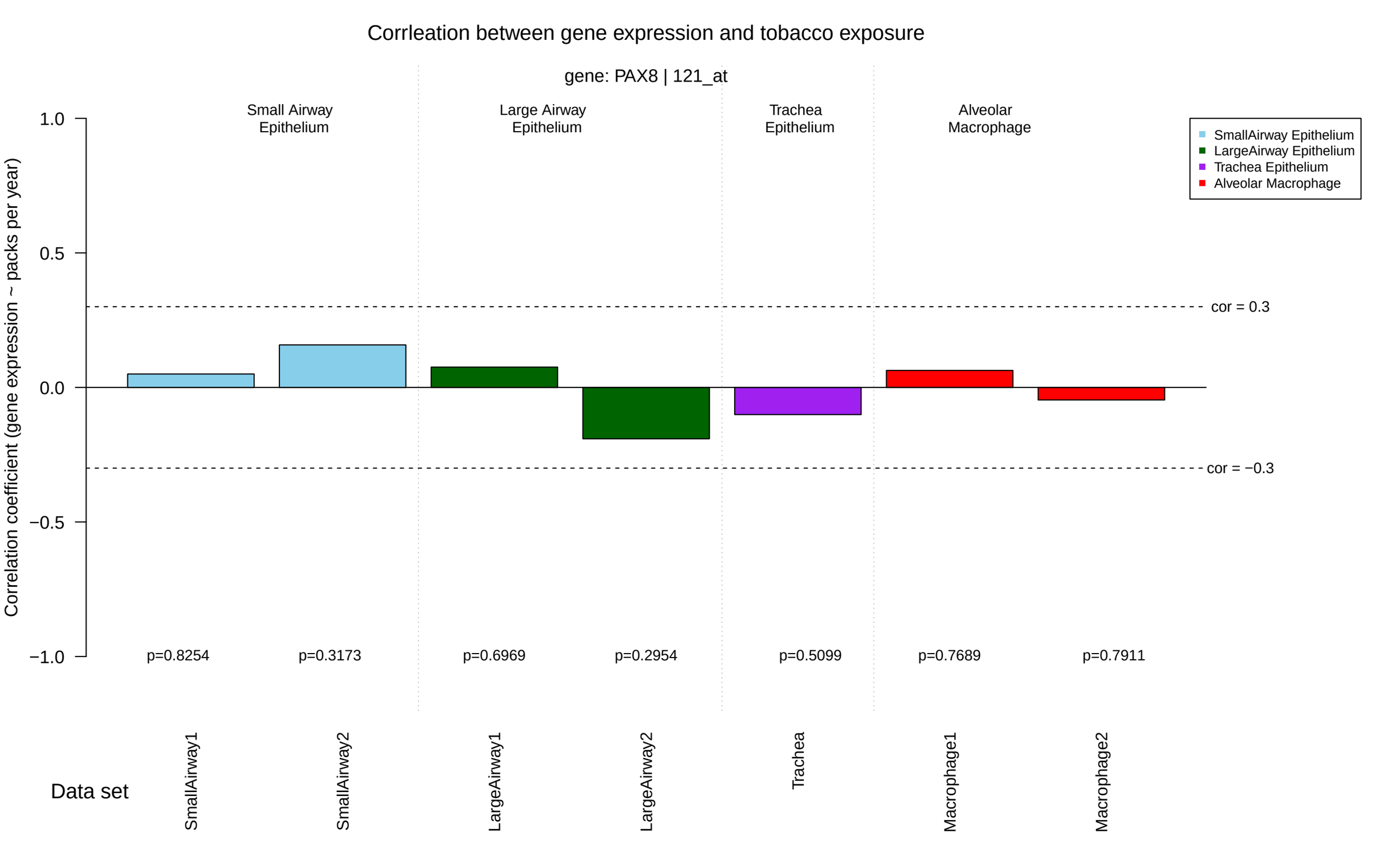

set: LargeAirway1 (p=0.2659)
PAX8 | 121_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4986)
PAX8 | 121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9008)
PAX8 | 121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4452)
PAX8 | 121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7834)
PAX8 | 121_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX8 | 121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8254
p=0.3173
p=0.6969
p=0.2954
p=0.5099
p=0.7689
p=0.7911
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2